Lab 7: Fuzzing for Input Validation Bugs

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Step 2: Running the buggy application (python recycling_rewards.py)

```
==== Running parse_collection ====
Enter material:weight in kg (e.g. paper:12): paper:12
Material and Weight: ('Paper', 12.0)

==== Running weight_average ====
Enter weights in kg as CSV (e.g. 5,12,20): 5,12,20
Average Weight: 12

==== Running net_points ====
Enter points|penalty (e.g. 80|10): 80|10
Net Points: 150.0

amanr@AOMEN-16 UCRT64 ~/Downloads/lab7-repo/Lab 7 (main)
$
```

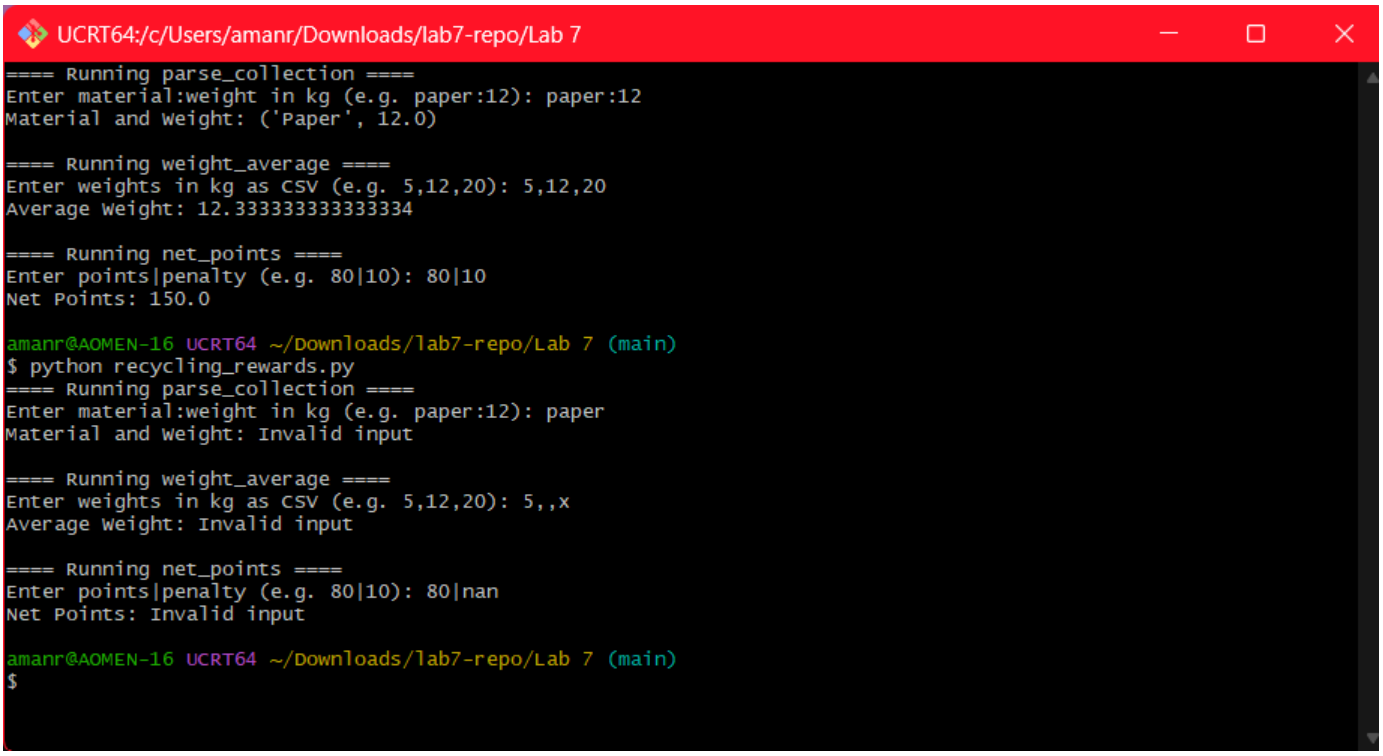
Step 4: Hypothesis fuzz tests failing on the buggy code (7 failed, 1 passed)

```
UCRT64:/c/Users/amanr/Downloads/lab7-repo/Lab 7
E                   '',
E               ),
E           )
E           Explanation:
E               These lines were always and only run by failing test cases:
E                   C:\Users\amanr\Downloads\lab7-repo\Lab 7\test_recycling_rewards.py:78

test_recycling_rewards.py:79: AssertionError
=========================== short test summary info ============================
FAILED test_recycling_rewards.py::test_parse_collection_no_crash - ExceptionGroup: Hypothesis found
3 distinct failures. (3 sub-exceptions)
FAILED test_recycling_rewards.py::test_parse_collection_valid_output - ExceptionGroup: Hypothesis fo
und 4 distinct failures. (4 sub-exceptions)
FAILED test_recycling_rewards.py::test_weight_average_no_crash - ExceptionGroup: Hypothesis found 2
distinct failures. (2 sub-exceptions)
FAILED test_recycling_rewards.py::test_weight_average_valid_output - AssertionError: weight_average
crashed with: ValueError("invalid literal for int() with base 10...
FAILED test_recycling_rewards.py::test_weight_average_is_correct - AssertionError: average of [0, 1]
 should be 0.5, got 0
FAILED test_recycling_rewards.py::test_net_points_no_crash - ExceptionGroup: Hypothesis found 3 dist
inct failures. (3 sub-exceptions)
FAILED test_recycling_rewards.py::test_net_points_valid_output - AssertionError: net_points crashed
with: ValueError("could not convert string to float: ''")
========================== 7 failed, 1 passed in 8.59s ==========================

amanr@AOMEN-16 UCRT64 ~/Downloads/lab7-repo/Lab 7 (main)
$
```

Step 5: Running the fixed application (valid and invalid inputs)



Step 6: Fuzz tests passing on the fixed code (8 passed)

